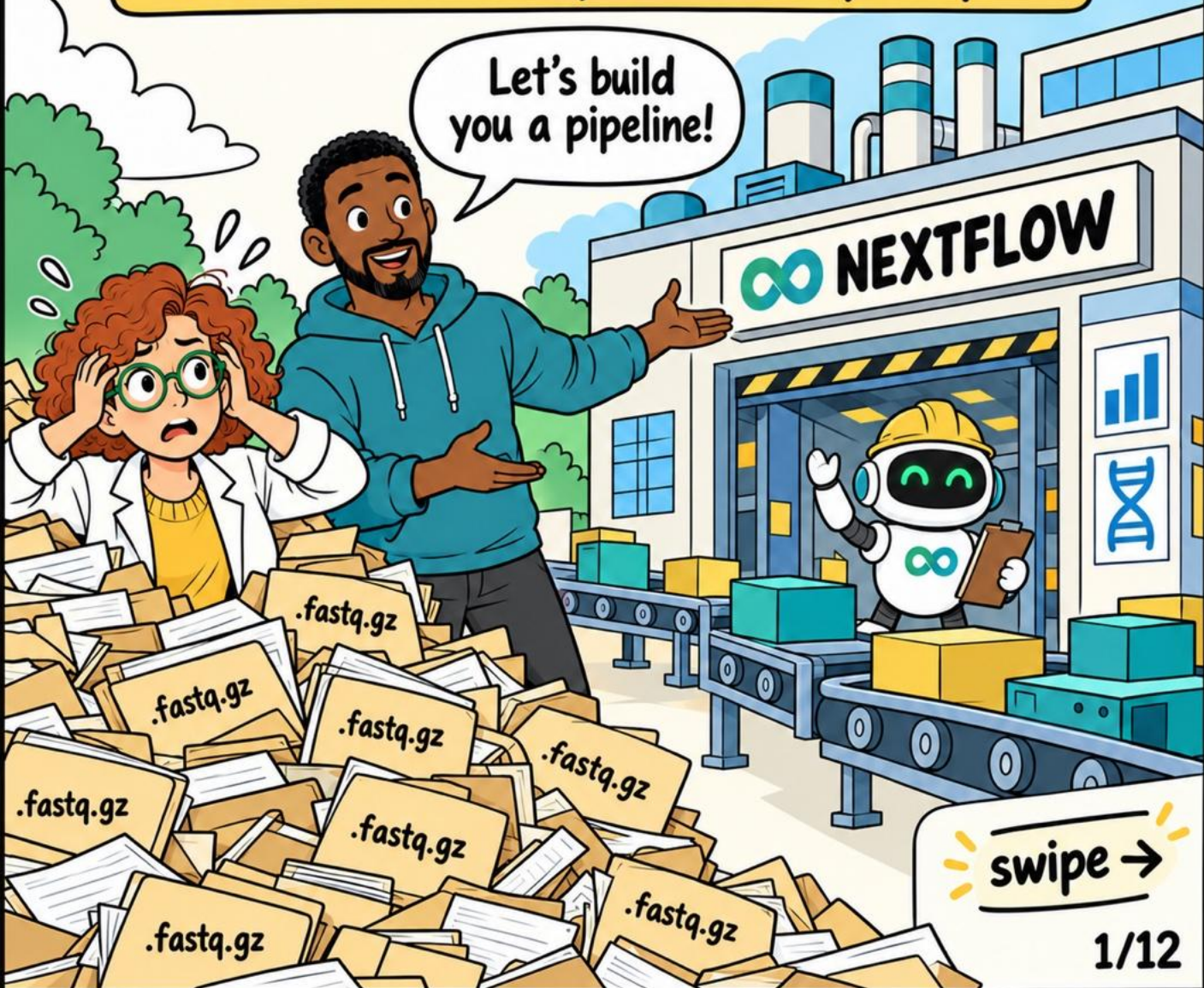


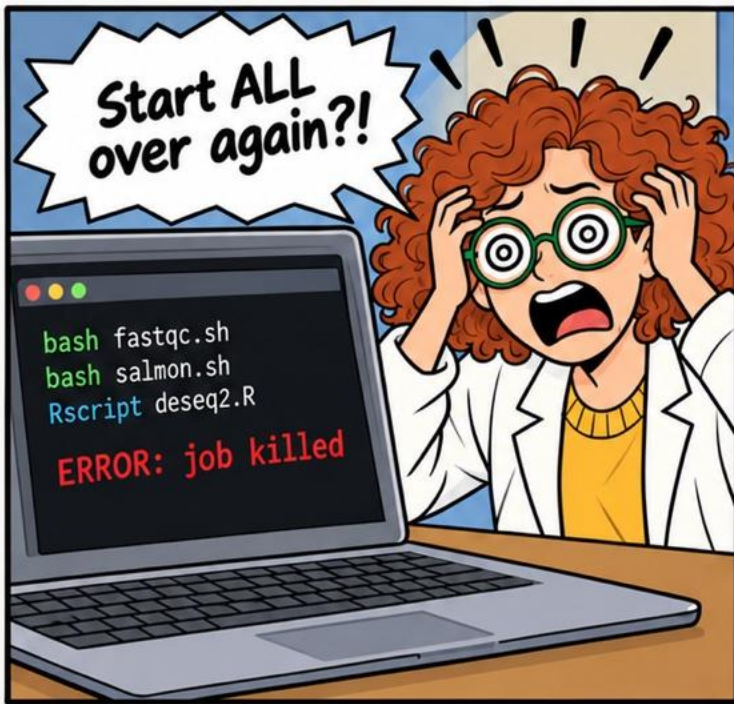
THE PIPELINE FACTORY

How Nextflow runs your RNA-seq analysis

Let's build you a pipeline!



THE SCRIPT CHAOS



WHAT'S IN THE FOLDER?

`my-rnaseq/`

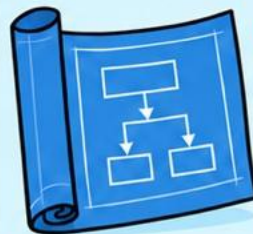
├─ `main.nf`

├─ `nextflow.config`

└─ `bin/`

├─ `run_salmon.sh`

└─ `deseq2.R`



main.nf =
the **BLUEPRINT**
(what runs,
in which order)



nextflow.config =
the **CONTROL ROOM**
(where & with what
resources)



bin/ =
YOUR TOOLBOX
(plain Bash & R scripts)

My old scripts
just go into **bin/**!



THE BIG IDEA: A FACTORY

PROCESS = one step
(FastQC, Salmon, DESeq2)

NEXTFLOW = the foreman
who starts each step as soon
as its input arrives

Salmon

Salmon

Salmon

S1

S2

S3

CHANNEL = an asynchronous queue
between steps (more on this soon!)

So all my samples
run in parallel...
automatically?

Yep!

WHAT IS A CHANNEL, REALLY?

An asynchronous FIFO queue that connects processes

PRODUCER

(a factory or a process)

CONSUMER

(QUANT)



nobody waits
for the whole batch

first in

first out

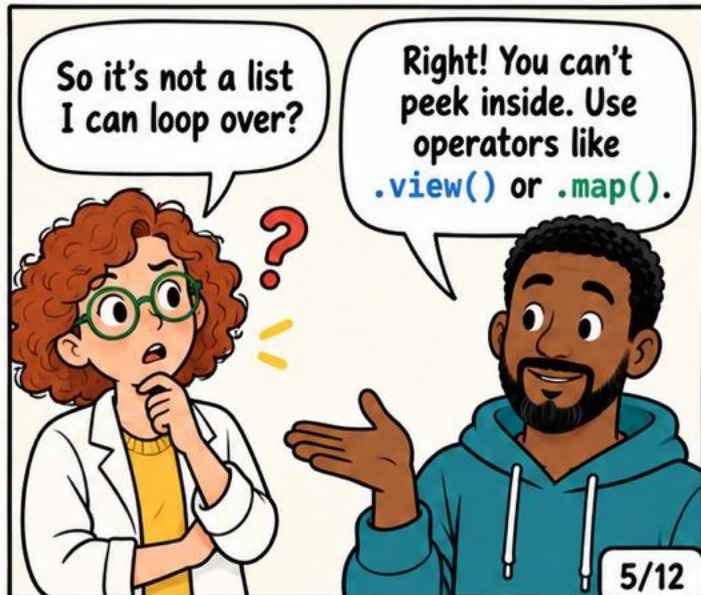
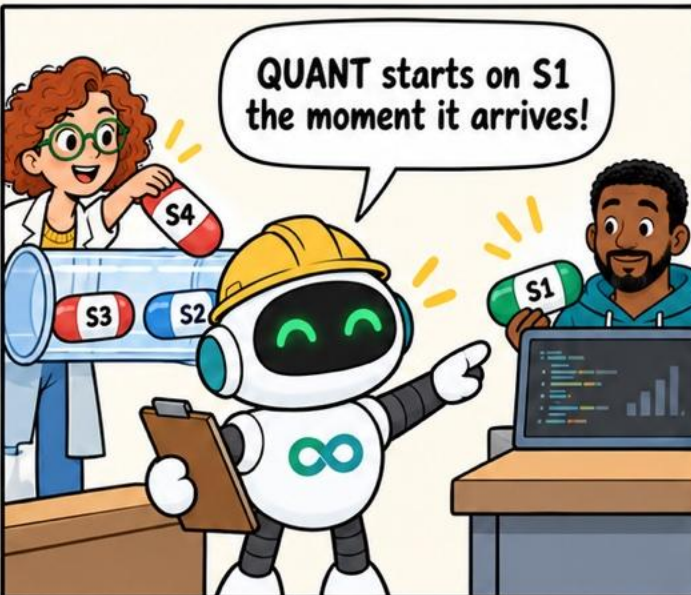
one direction



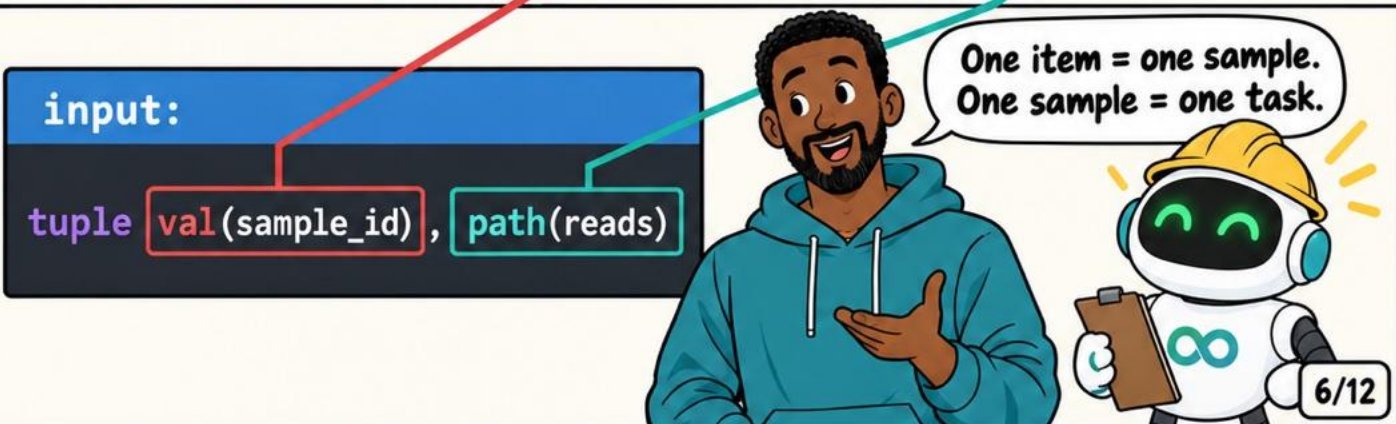
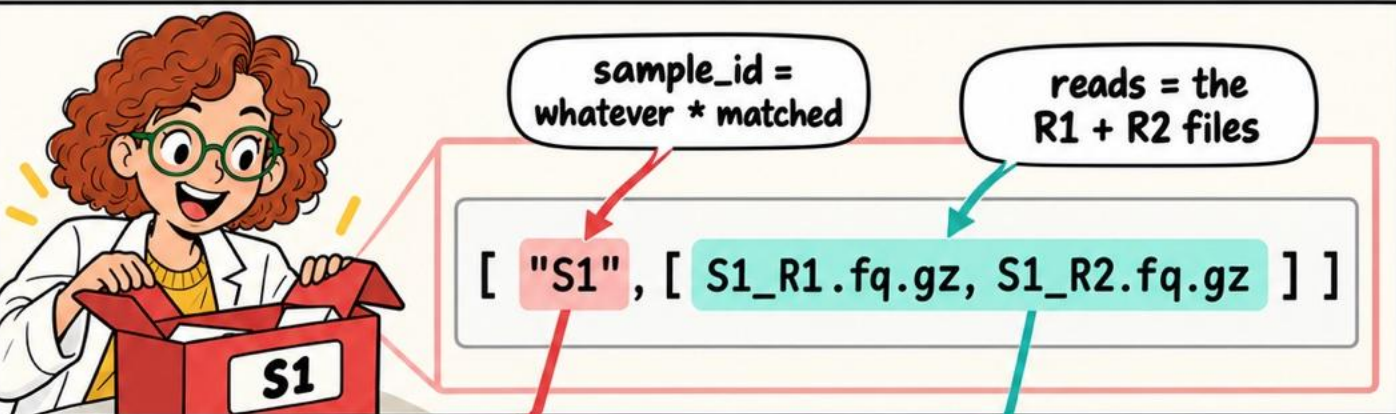
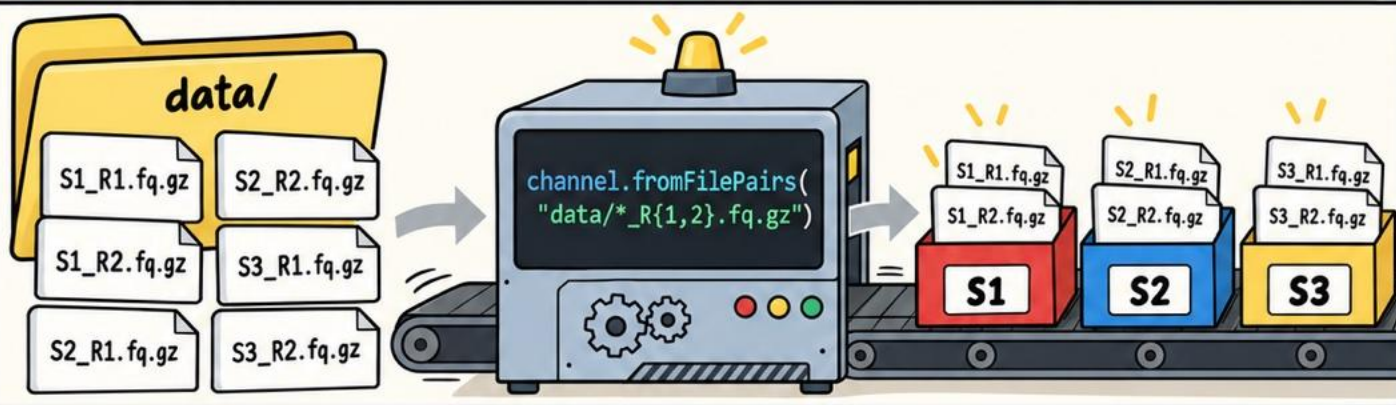
QUANT starts on S1
the moment it arrives!

So it's not a list
I can loop over?

Right! You can't
peek inside. Use
operators like
`.view()` or `.map()`.



WHAT IS 'S1'? OPENING ONE ITEM



TWO KINDS OF CHANNEL

QUEUE CHANNEL

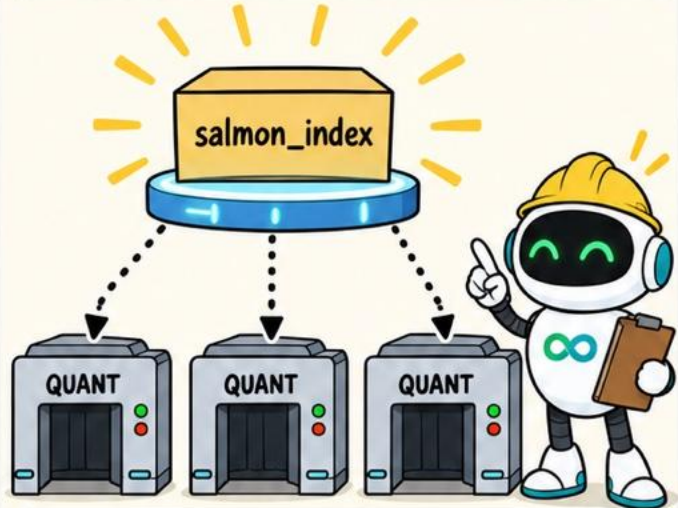
```
channel.of("S1", "S2", "S3")
```



Many items. Each one is consumed once.
One task per item.

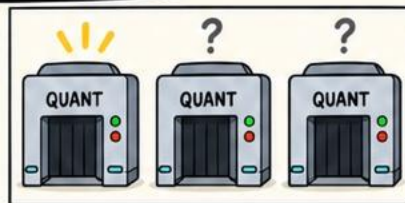
VALUE CHANNEL

```
channel.value("salmon_index")
```



One item. Reused by every task, forever.

Why did QUANT
only run once?!



Your index was
a queue channel.
Make it a value
channel!

CHANNEL FACTORIES: WHERE ITEMS COME FROM

```
channel.of(1, 2, 3)
```

queue channel



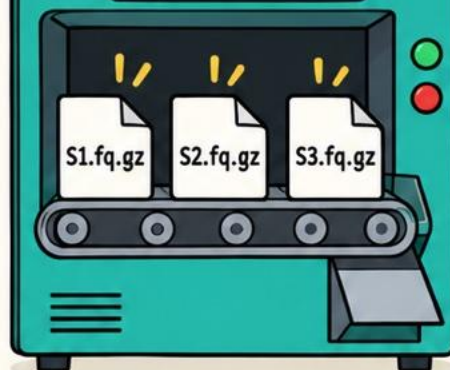
```
channel.fromList(['salmon', 'kallisto'])
```

queue channel



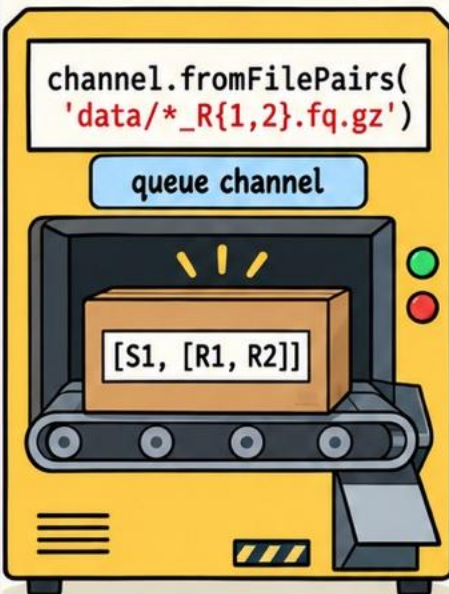
```
channel.fromPath('data/*.fq.gz')
```

queue channel



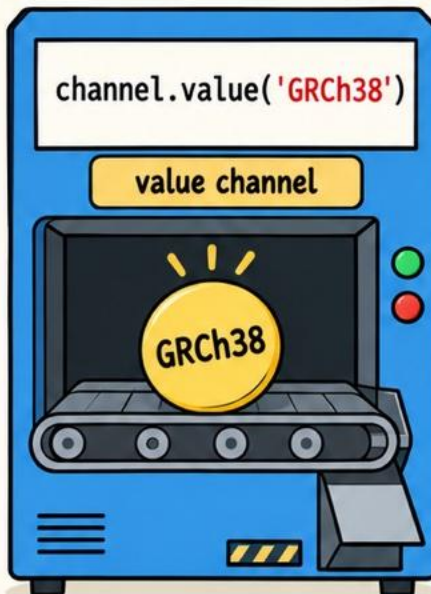
```
channel.fromFilePairs('data/*_R{1,2}.fq.gz')
```

queue channel

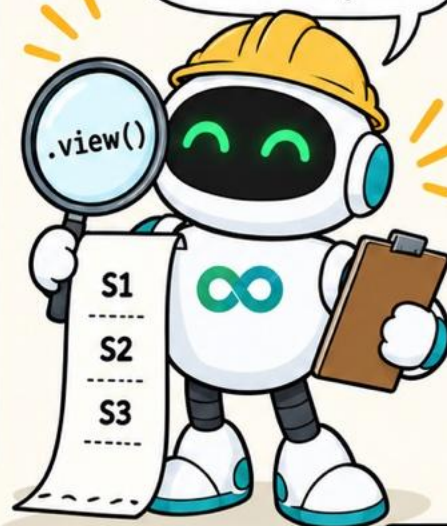


```
channel.value('GRCh38')
```

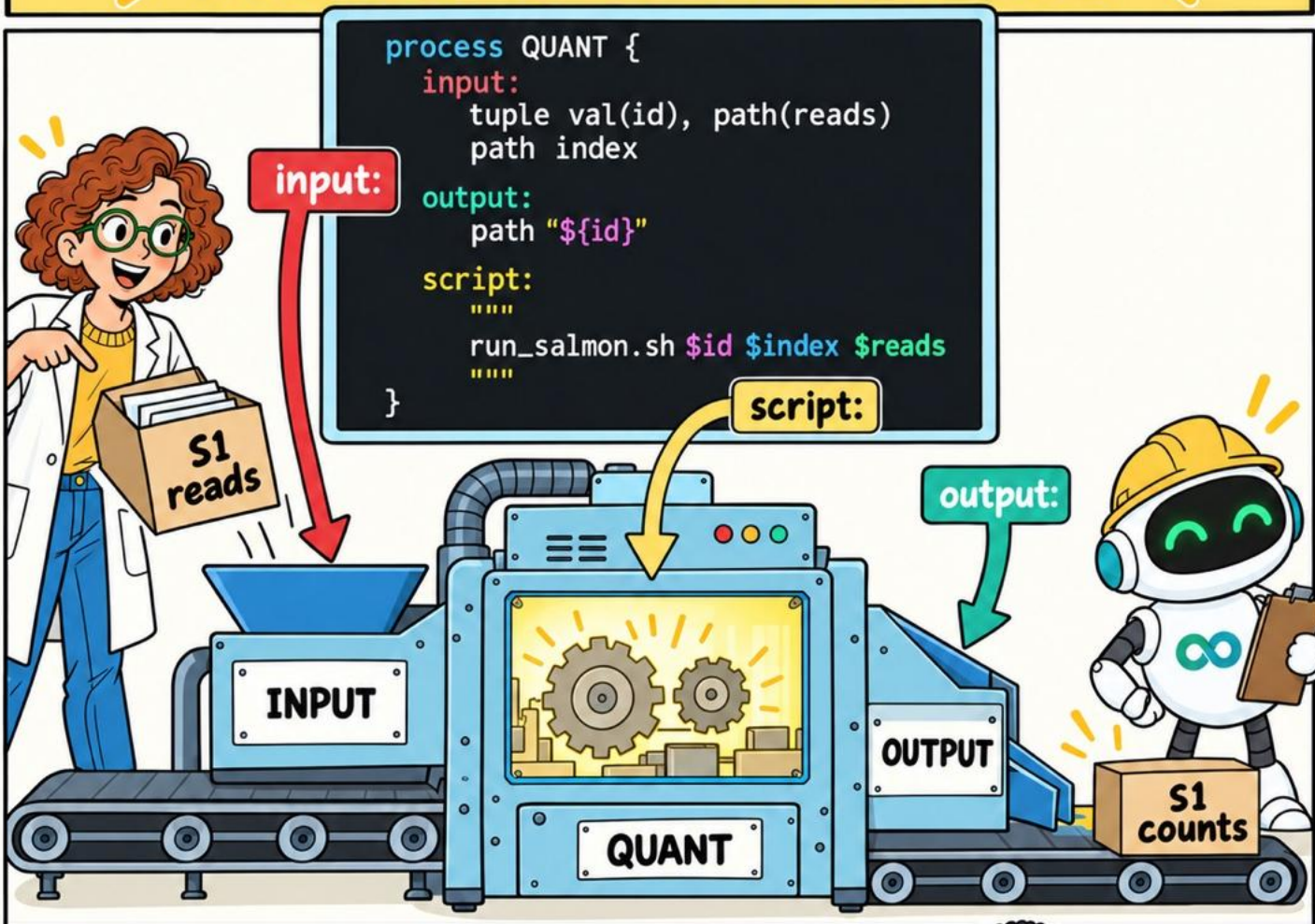
value channel



Add `.view()` to see every item!



ANATOMY OF A PROCESS

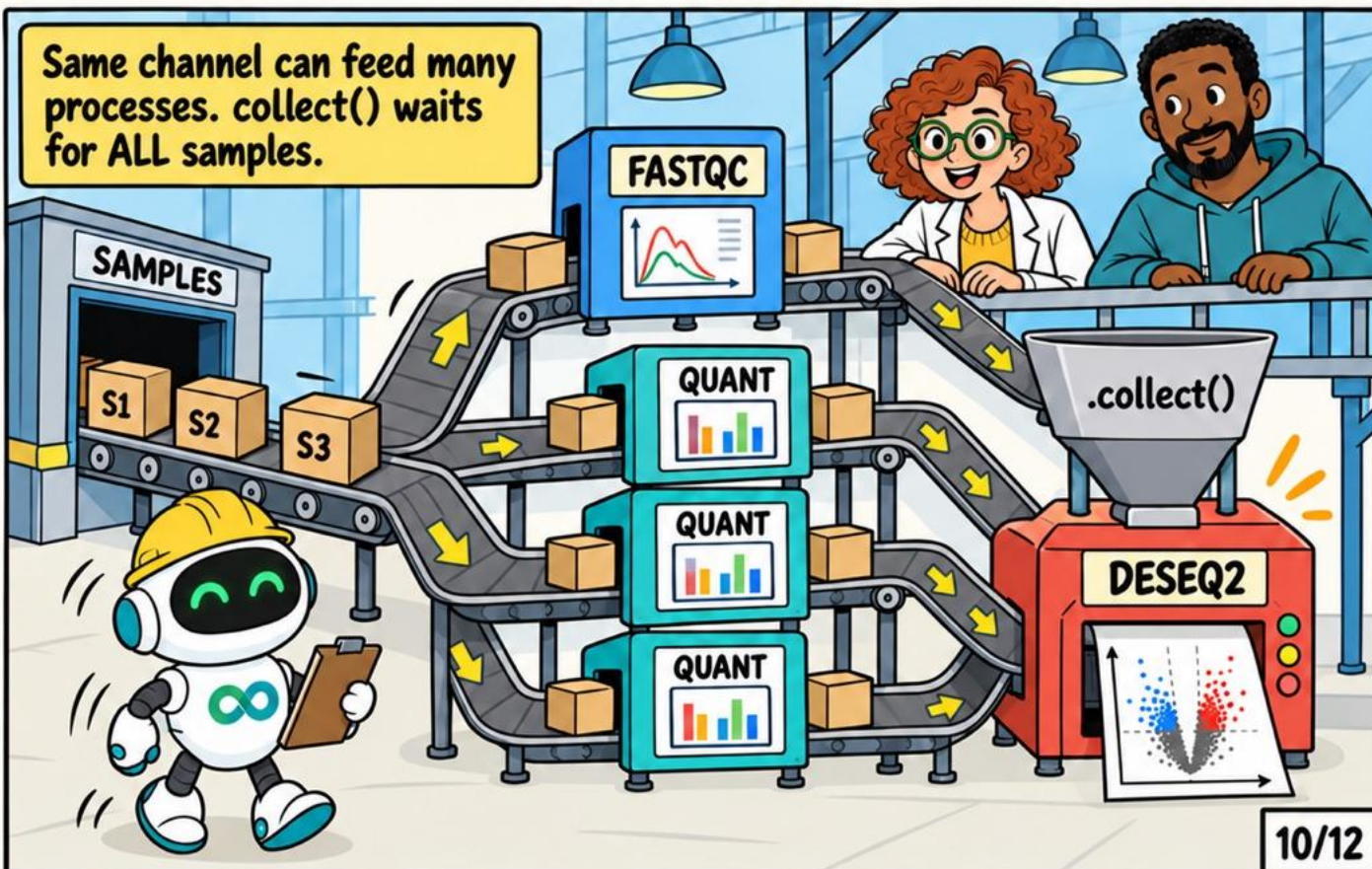


Each task runs in its own clean folder: work/ab/12cd...

main.nf: WIRING THE BELTS

```
workflow {  
  reads = channel.fromFilePairs("data/*_R{1,2}.fq.gz")  
  FASTQC(reads)  
  QUANT(reads, index)  
  DESEQ2(QUANT.out.collect())  
}
```

Same channel can feed many processes. `collect()` waits for ALL samples.



nextflow.config: THE CONTROL ROOM

So the config is
WHERE it runs,
main.nf is
WHAT runs!

Same main.nf.
Just change
the profile.

```
params.reads = "data/*_R{1,2}.fq.gz"  
process.cpus = 2  
  
withName: QUANT { cpus = 8 }  
  
profiles {  
  docker { docker.enabled = true }  
  slurm { process.executor = 'slurm' }  
}
```

PIPELINES
POWER
DISCOVERY



SCALE
FASTER
SCIENCE

FROM LAPTOP
TO CLUSTER

LAPTOP

DOCKER

HPC
CLUSTER



➤ RUN IT. CRASH? RESUME! ➤

